CODING-DECODING

Verbal Reasoning | Section-I: General Mental Ability | Chapter-4

Complete Concepts, Shortcuts & Practice Problems

Based on: S Chand Academy – Coding-Decoding | PART-I

LEARNING OBJECTIVES

Understand what Coding-Decoding is and why it matters in competitive exams

Master all major types: Letter Shifting, Reverse Order, Opposite Letters, Number Coding, Symbol Coding, Mixed Coding

Learn shortcut methods to solve problems in under 30 seconds

Apply step-by-step decoding strategies to SSC, Banking & Railway exam questions

Build confidence through solved examples and practice quizzes

WHAT IS CODING-DECODING?

Definition: Coding-Decoding is a method of transmitting a message between sender and receiver such that no third person can understand it. In exams, a word/number is coded using a specific pattern, and candidates must decode it.

Real-World Analogy

Just like Morse Code (••• — — — ••• = SOS)

Just like Barcodes on products

Just like encrypted WhatsApp messages

In exams: A pattern hides the real word!

Exam Relevance

SSC CGL/CHSL: 2-4 questions

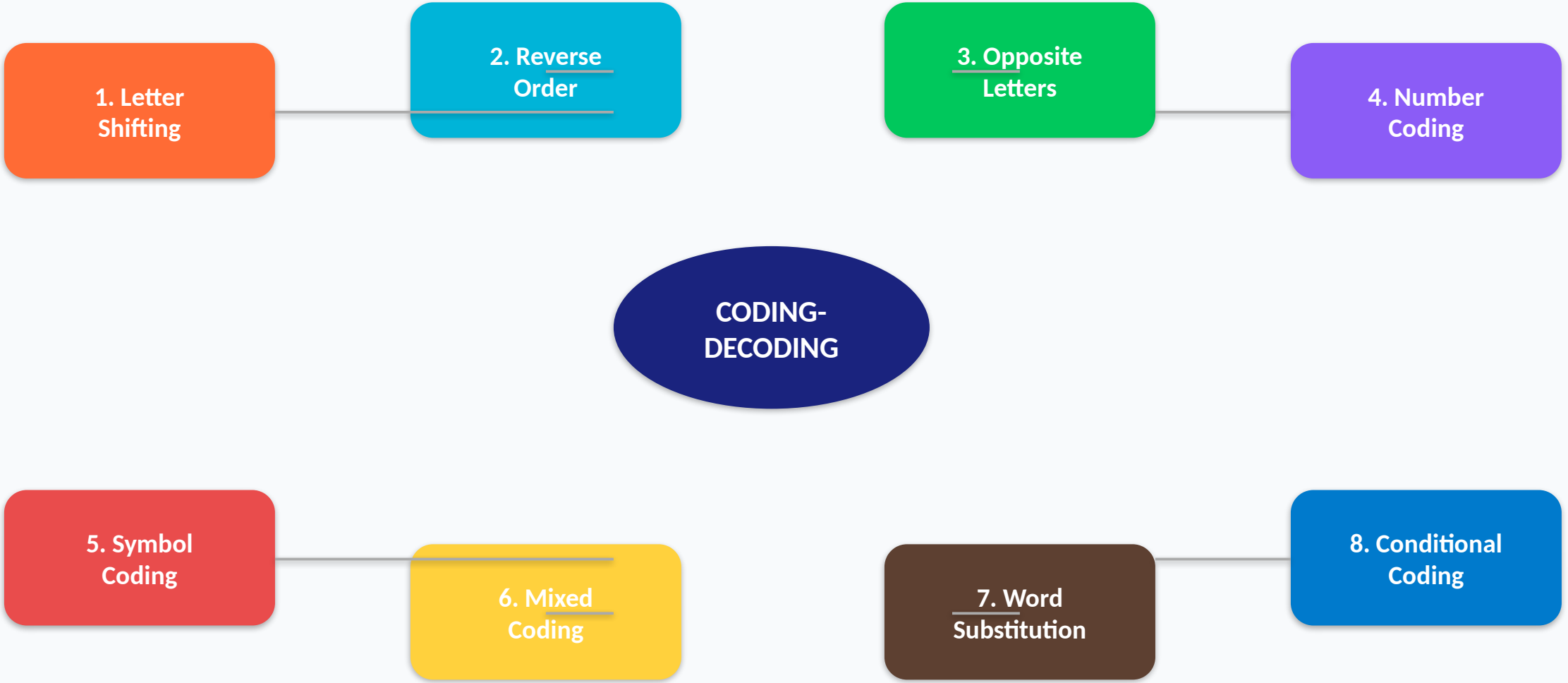
Bank PO/Clerk: 3-5 questions

Railways NTPC/Group D: 2-3 questions

UPSC CSAT: 1-2 questions

Time per question: 30-45 seconds

TYPES OF CODING-DECODING (MIND MAP)



TYPE 1: LETTER SHIFTING (+1, +2, -1, -2, etc.)

Concept: Each letter of the word is shifted forward or backward in the alphabet by a fixed number.

Analogy: Like turning a combination lock — each dial moves by the same number of clicks!

Example 1: +1 Pattern

If CAT is coded as DBU, how is DOG coded?

Solution: C→D (+1), A→B (+1), T→U (+1)

So DOG → EPH

Example 2: +2, -1 Pattern

If MATH is coded as OCUG, find the code for CODE?

Pattern: +2, +2, -1, -1

CODE → EQDD

SHORTCUT: Write the alphabet with positions (A=1 to Z=26). For quick +1/-1, use the alphabet circle. For mixed patterns, write position numbers below each letter first.

Pro Tip: If the shift is +13, it's the same as opposite letter! (A ↔ N, B ↔ O)

TYPE 2: REVERSE ORDER

Concept: The letters of the word are written in reverse (backward) order.

Analogy: Like reading a word in a mirror — it looks backwards!

Example 1: Simple Reversal

If ROSE is coded as ESOR, how is LOTUS coded?

Solution: R-O-S-E → E-S-O-R (reversed)

So LOTUS → SUTOL

Example 2: Decoding Reverse

If FLOWER is coded as REWOLF, decode MVIP?

Solution: Reverse the code → PVIM

Answer: PVIM

TRAP ALERT: Sometimes reversal is combined with shifting!

Always test both pure reversal AND reversal + shift patterns.

Shortcut: If code starts with original word's LAST letter → likely simple reversal.

TYPE 3: OPPOSITE LETTERS

Concept: Each letter is replaced by its opposite letter in the alphabet (A ↔ Z, B ↔ Y, C ↔ X, ...).

Analogy: Like two ends of a seesaw — when one goes up, the other goes down!

Opposite Letter Chart

A	B	C	D	E	F	G	H	I	J	K	L	M
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
Z	Y	X	W	V	U	T	S	R	Q	P	O	N

Example 1: Opposite Coding

If GOOD is coded as TLLW, how is BAD coded?

Solution: G ↔ T, O ↔ L, O ↔ L, D ↔ W

BAD → YZW

SHORTCUT: Sum of opposite letter positions = 27 (A=1, Z=26 → 1+26=27). If you know one, subtract from 27 to get the other. Memorize the first 5 pairs: A-Z, B-Y, C-X, D-W, E-V — the rest follow automatically.

Memory Hook: 'AZ BY CX DW EV FU GT HS IR JQ KP LO MN'

TYPE 4: NUMBER CODING

Concept: Letters are replaced by their position numbers (A=1, B=2... Z=26) or by modified numbers.

Analogy: Like a telephone keypad — each button represents a letter or number!

Example 1: Direct Position

If CAT is coded as 3120, how is DOG coded?

Solution: C=3, A=1, T=20 → 3120

DOG → 4157 (D=4, O=15, G=7)

Example 2: Sum of Positions

If BAD = 6, and CAT = 24, how is RAT coded?

Solution: B+A+D = 2+1+4 = 7... Wait!

Pattern: Sum of positions. CAT = 3+1+20 = 24 ✓

RAT = 18+1+20 = 39

SHORTCUT: Always check three patterns first: (1) Direct position concatenation, (2) Sum of positions, (3) Product of positions. If none work, try sum of digits of positions or square of positions.

Pro Tip: Write A=1, B=2... Z=26 on your rough sheet before starting the exam — it saves 2 minutes!

TYPE 5: SYMBOL CODING

Concept: Letters/words are replaced by symbols (@, #, $, %, *, etc.) based on specific rules.

Analogy: Like emojis replacing words in a text message!

Example 1: Letter-Symbol Map

If A is @, B is #, C is $, how is CAB coded?

Solution: C=$, A=@, B=#

CAB → $@#

Example 2: Word-Symbol Logic

'He is smart' = '@ # $'

'She is kind' = '% # &'

'He and She' = '@ * %'

What is the code for 'and'?

Answer: * (by cross-matching)

SHORTCUT: For word-symbol coding, make a table. Cross-match sentences to find common words and their symbols. Eliminate known words to isolate unknowns.

Exam Strategy: These questions look scary but are 100% logical. Don't skip them!

TYPE 6: MIXED CODING

Concept: Two or more coding types are combined (e.g., Reverse + Shift, Opposite + Number).

Analogy: Like a recipe with multiple spices — you must identify each ingredient!

Example 1: Reverse + Shift

If FLOWER is coded as SFMPXF, decode the pattern.

Step 1: F→S? No direct shift. Try reverse: FLOWER → REWOLF

Step 2: R→S (+1), E→F (+1), W→X (+1)... Yes!

Pattern: Reverse word, then +1 each letter

Example 2: Opposite + Number

If A=26, B=25, C=24... (opposite number), how is AZ coded?

A = 26, Z = 1

AZ → 261

SHORTCUT – PEEL THE ONION: (1) Check simple reversal first, (2) Check shifting (+1, +2, -1), (3) Check opposite letters, (4) Combine findings. Golden Rule: Most mixed codes use ONLY 2 operations — don't overcomplicate!

TYPE 7: WORD SUBSTITUTION

Concept: Whole words are replaced by other words (not letters).

Analogy: Like slang — 'cool' means 'good,' 'lit' means 'exciting'!

Example 1: Word Chain

If 'book' is called 'pen', 'pen' is called 'paper', 'paper' is called 'eraser', what do you write on?
Solution: You write on 'paper' — but 'paper' is called 'eraser'.
Answer: eraser

Example 2: Color Substitution

If 'red' means 'blue', 'blue' means 'green', 'green' means 'yellow', what is the color of sky?
Solution: Sky is blue. But 'blue' means 'green'.
Answer: green

TRAP ALERT: NEVER use real-world logic — only use the given code!

The sky is blue in reality, but in this code world, blue = green. Follow the chain strictly. Read the question twice!

TYPE 8: CONDITIONAL CODING

Concept: Coding rules change based on conditions (vowel/consonant, first letter, word length, etc.).

Analogy: Like traffic rules — red means stop, green means go. The action depends on the condition!

Example 1: Vowel/Consonant

If first letter is vowel → code = +1. If consonant → code = -1.
Code for APPLE and BAT.
APPLE: A is vowel → +1 each → BQQMF
BAT: B is consonant → -1 each → AZS

Example 2: Word Length

If word has 4 letters → reverse it. If 5 letters → +2 each.
Code for ROSE and FLORA.
ROSE (4 letters) → ESOR
FLORA (5 letters) → HNQTC

SHORTCUT: ALWAYS identify the condition first before applying any code. Underline condition words ('if vowel,' 'if length > 4'). Create a quick decision tree on rough paper.

Time Saver: If multiple words are given, group them by condition first, then code in batches.

MASTER SHORTCUT TRICKS

Alphabet Number Line	Always write A=1 to Z=26 first
27 Rule	For opposite letters (A+Z=27)
PEEL THE ONION	For mixed/hybrid codes
Cross-Match Table	For symbol/word substitution
Batch Processing	For conditional coding
Reverse Test	First check when code looks 'jumbled'
Position Difference	When shifts are irregular

In Coding-Decoding, the pattern is always simpler than it looks. Don't overthink!

PRACTICE QUIZ (5 Questions)

Q1. If TABLE is coded as UBCMF, how is CHAIR coded?

Q2. If SCHOOL is coded as LOOHCS, how is CLASS coded?

Q3. If LIGHT is coded as ORTSG, what coding is used?

Q4. If BAD = 214, and CAT = 3120, how is RAT coded?

Q5. If 'pen' means 'book', 'book' means 'bag', what do you read?

Time Limit: 5 Minutes

COMMON MISTAKES TO AVOID

Ignoring the alphabet position chart → Always write A-Z with numbers first

Using real-world meaning in word substitution → Trust ONLY the given code

Assuming only one operation in mixed coding → Test reversal + shift combinations

Rushing without identifying the condition → In conditional coding, underline the rule first

Confusing +1 with -1 direction → Write arrows: Forward → +, Backward → –

Skipping symbol coding questions → They are logical, not mathematical!

Every mistake is a pattern you haven't recognized yet. Slow down to speed up!

KEY TAKEAWAYS

Coding-Decoding is about pattern recognition, not memorization

Always write the A-Z position chart (A=1 to Z=26) before solving

Use the 'PEEL THE ONION' method for mixed codes

In word substitution, follow the code world, not the real world

Target time: 30-45 seconds per question after practice

Maintain a formula card with opposite letters and common shifts

"The code is not hiding from you — you just need to look at it from the right angle!"

RESOURCES & REFERENCES

Primary Book: A Modern Approach to Verbal & Non-Verbal Reasoning by Dr. R.S. Aggarwal (S Chand Publishing)

Video Source: S Chand Academy – Coding-Decoding | PART-I (YouTube)

Practice Platforms: Testbook, BYJU'S Exam Prep, Gradeup

Previous Papers: SSC CGL (Tier-I & II), SBI PO, IBPS Clerk, Railways NTPC

Mobile Apps: Adda247, Oliveboard, Pocket Aptitude

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